

HEALTH INSIGHT ANALYTICS

EMPOWERING CARE
THROUGH DATA
INSIGHTS



Data-Driven Insights for Healthcare Success

Introduction

- This dataset represents a **Healthcare Organization** that operates across multiple hospitals and delivers a wide range of medical. It encompasses comprehensive patient-level data. The dataset offers valuable insights into how the organization manages **patient care, treatment workflows and operational efficiency** across its various departments.



STEPS INVOLVED IN DATA ANALYSIS



DATA COLLECTION
AND LOADING



DATA OVERVIEW

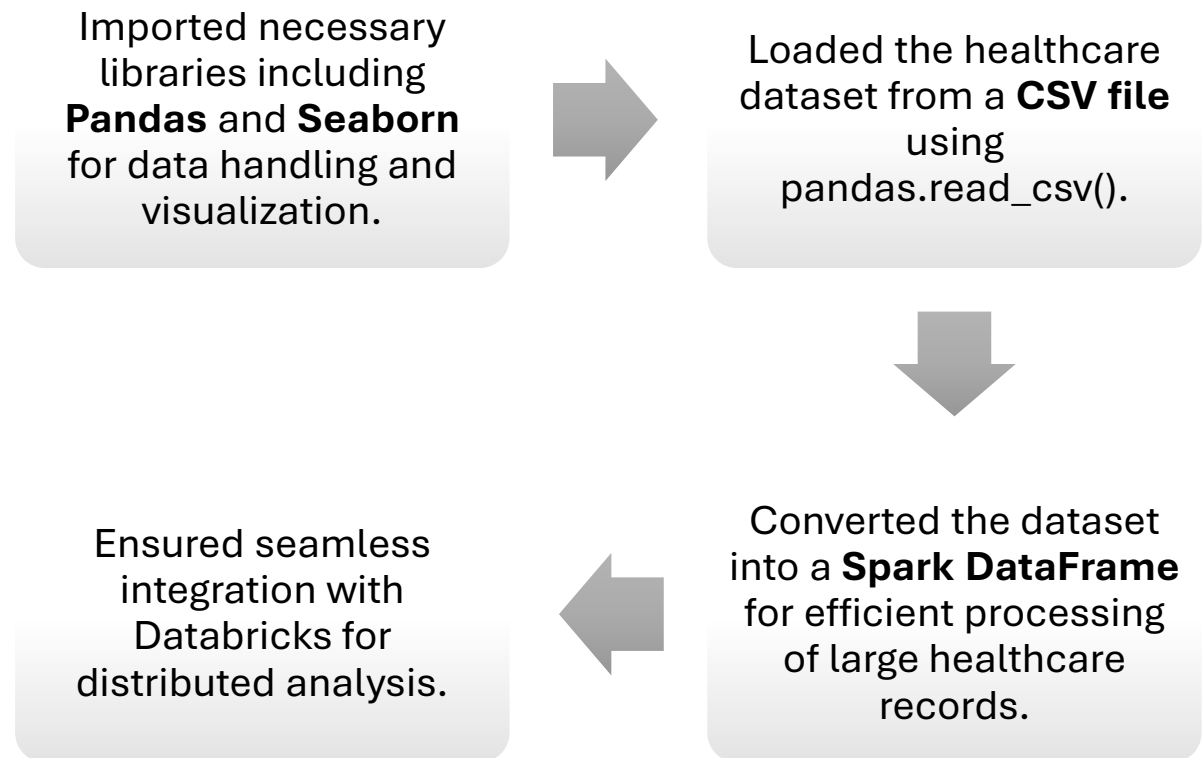


DATA CLEANING

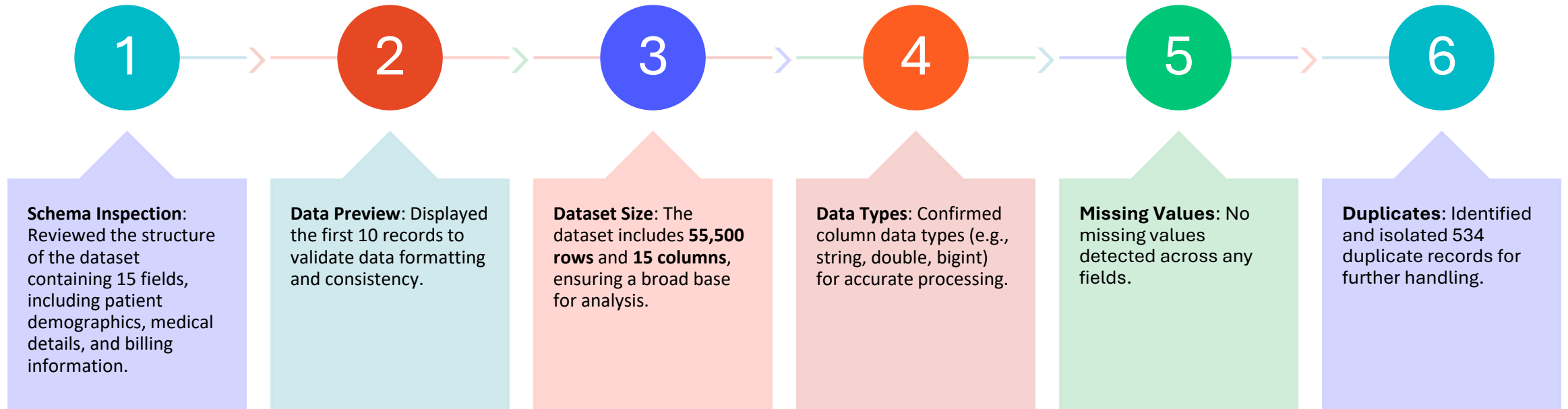


DATA ANALYSIS
AND INSIGHTS

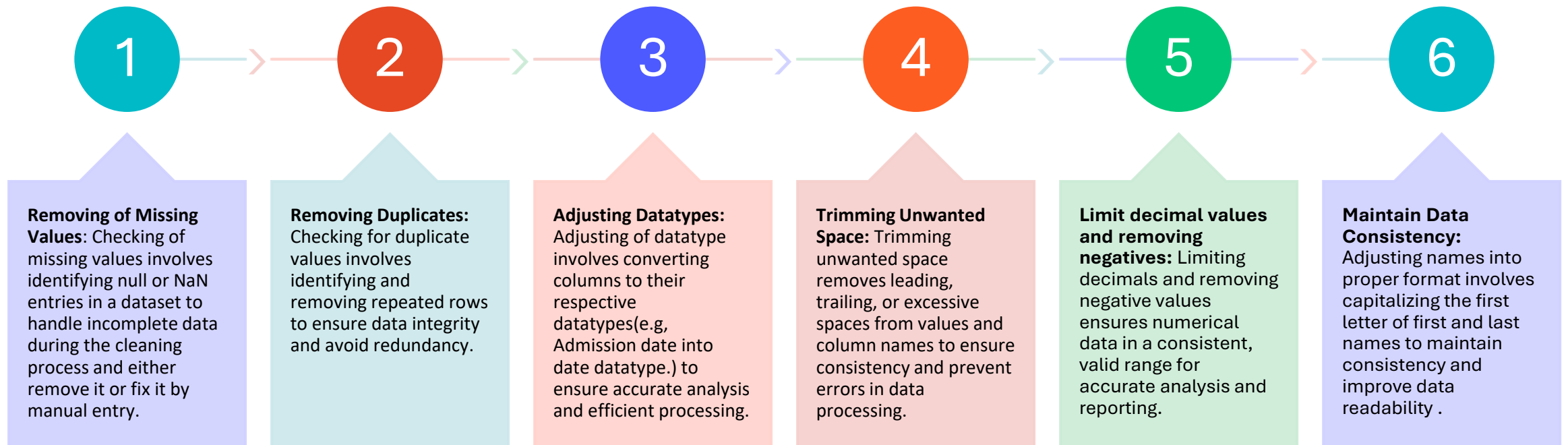
STEP 1 : DATA COLLECTION AND LOADING



STEP 2 : DATA OVERVIEW



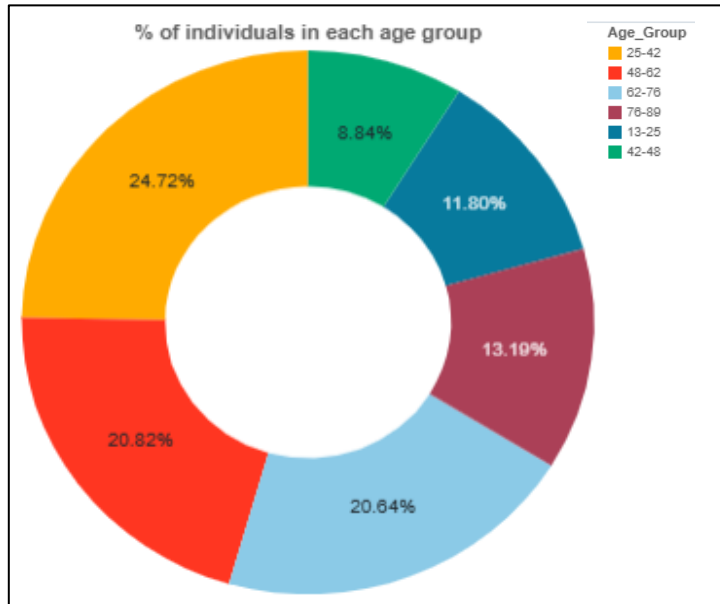
STEP 3 : DATA CLEANING



STEP 4: DATA ANALYSIS AND INSIGHTS

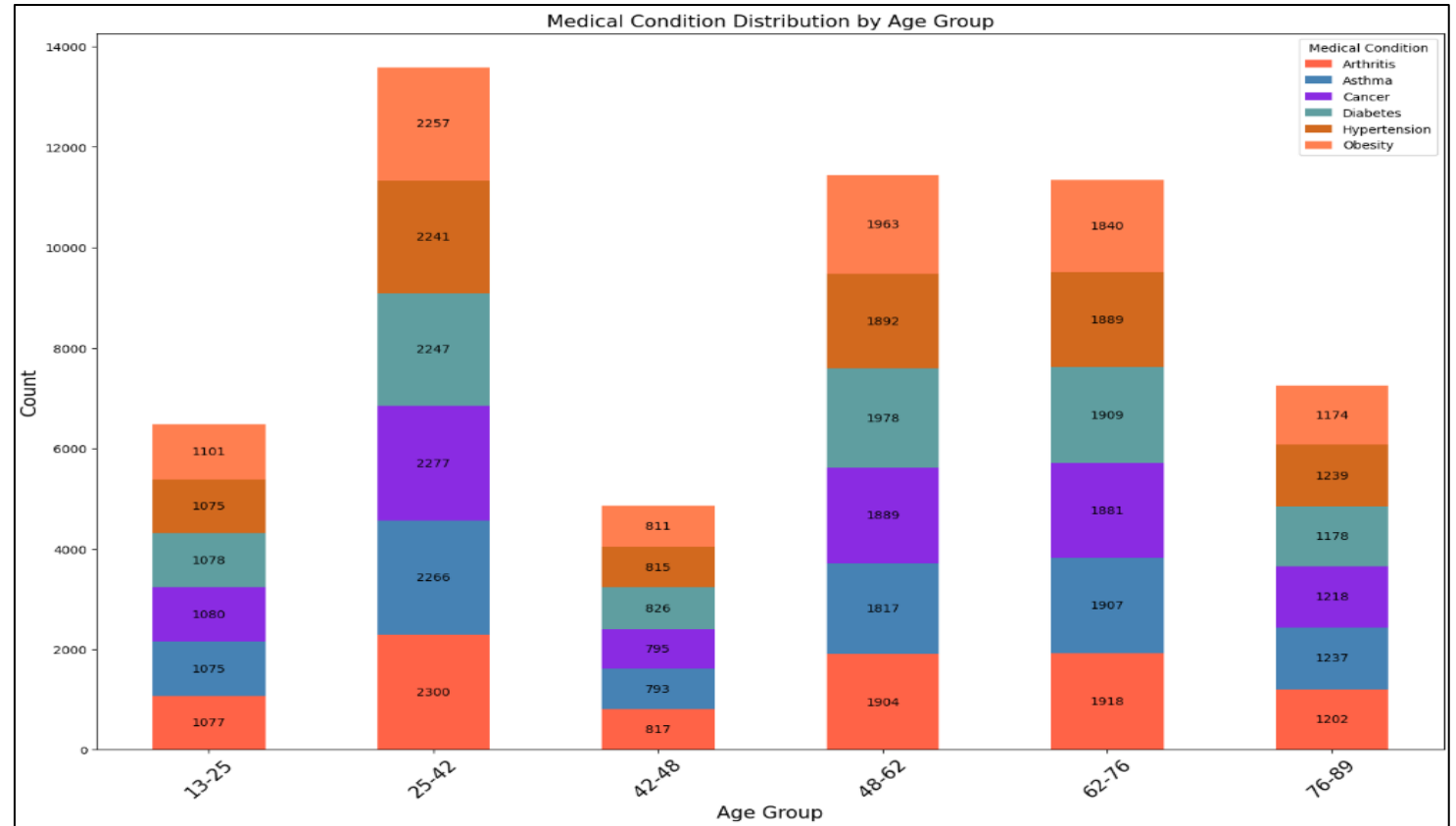


1. Demographics

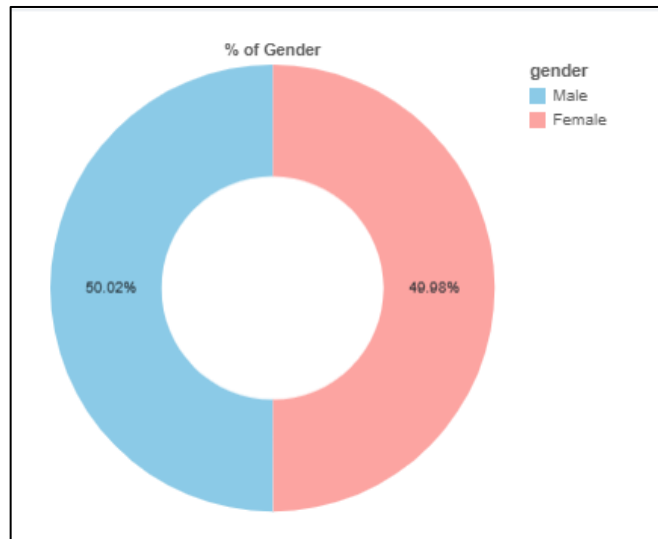


Age Distribution :

- The 25–42 age group forms the largest segment (24.72%), followed by older adults aged 48–76 at 20.82%, highlighting significant demand across healthcare services.
- The lower representation of 42–48 (8.84%) may indicate an underutilization of healthcare resources in that age range.

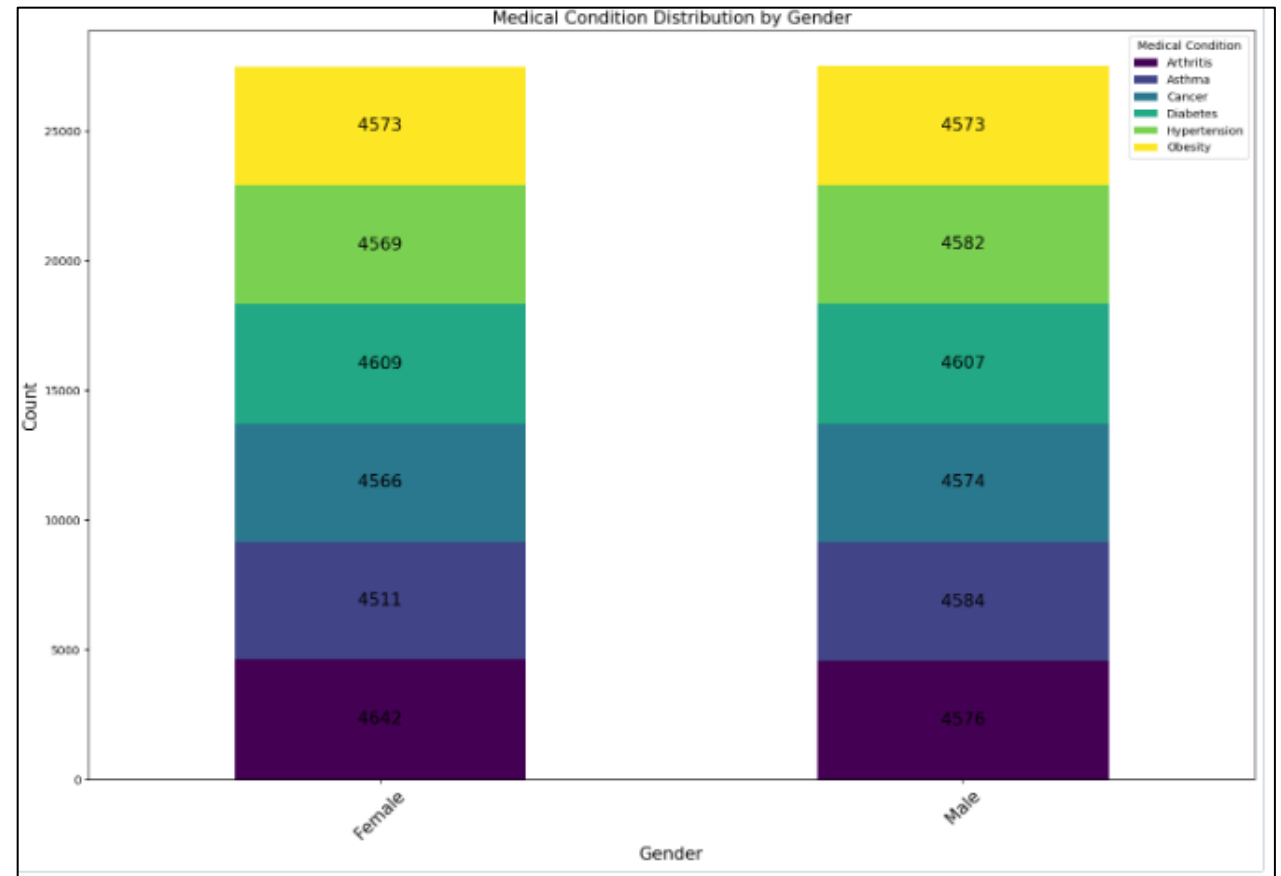


- Arthritis peaks in both middle-aged (25–42) and older adults (62–76), while diabetes is most common in the 48–62 range, reflecting age-related joint and metabolic issues.
- Obesity is highest among young adults (13–25) indicating early lifestyle risks, whereas hypertension is most prevalent in seniors (76–89), consistent with age-related vascular changes.



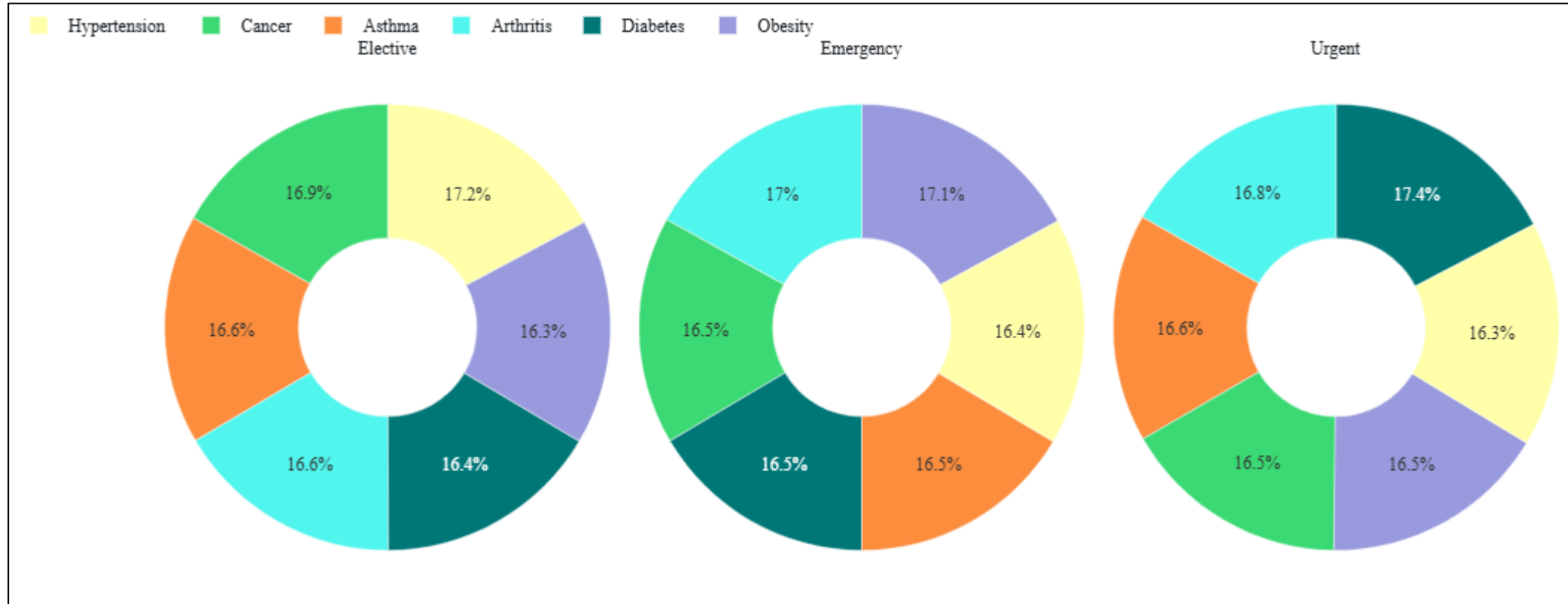
Gender distribution :

- The dataset shows a nearly equal gender distribution, with males at 50.02% and females at 49.98%.
- This equitable portrayal guarantees impartial perspectives regarding genders in subsequent healthcare evaluations.



- The distribution of medical conditions across genders is nearly identical, with all condition counts differing by less than 100 cases between males and females.
- Arthritis is slightly more common in females (4642 vs. 4576), while asthma, cancer, diabetes, hypertension and obesity show minimal gender-based variation, indicating balanced healthcare needs across both genders.

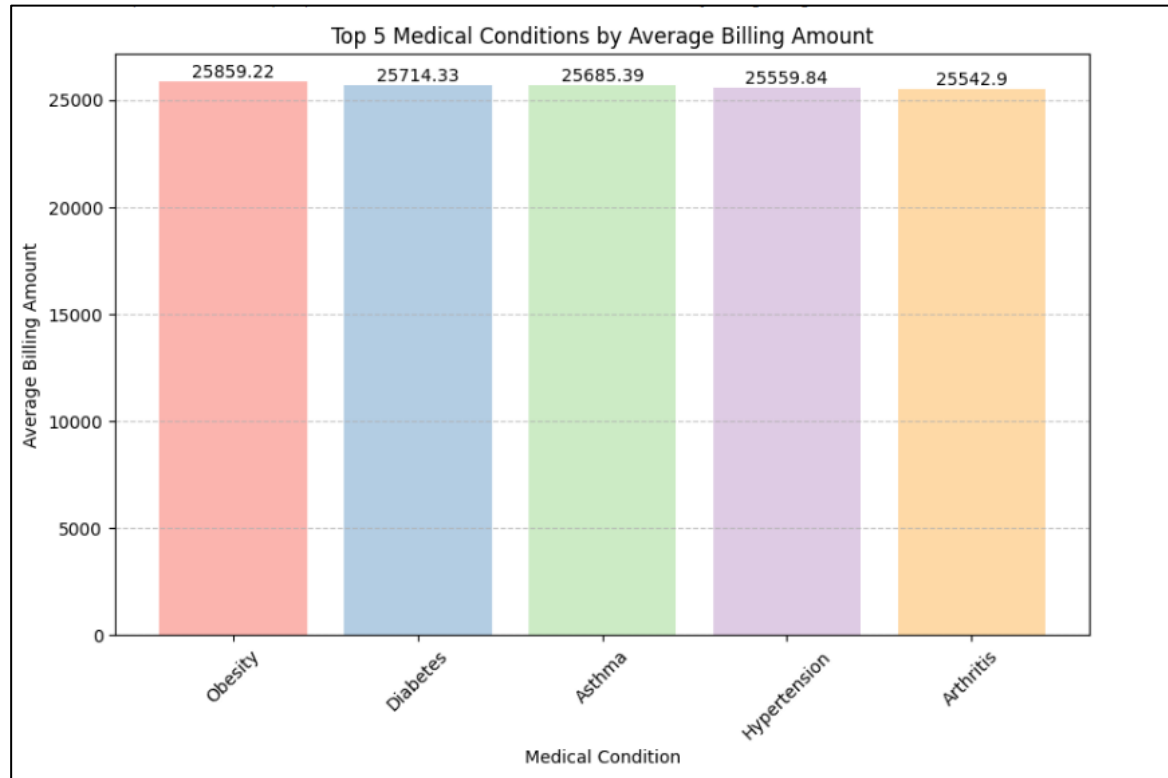
2. Hospitalisation Patterns



The admission type as per the medical conditions

- In Elective admissions most of the cases are related with Hypertension (17.2%), closely followed by Diabetes and Cancer, suggesting these ailments may require scheduled therapies or regular procedures.
- Emergency admissions are most frequent for Cancer (17%) and Asthma (16.6%), reflecting their potential to lead to sudden, severe episodes requiring immediate care.
- Urgent admissions are primarily driven by Obesity (17.4%), indicating that complications associated with this issue frequently require prompt, though not necessarily immediate, treatment.

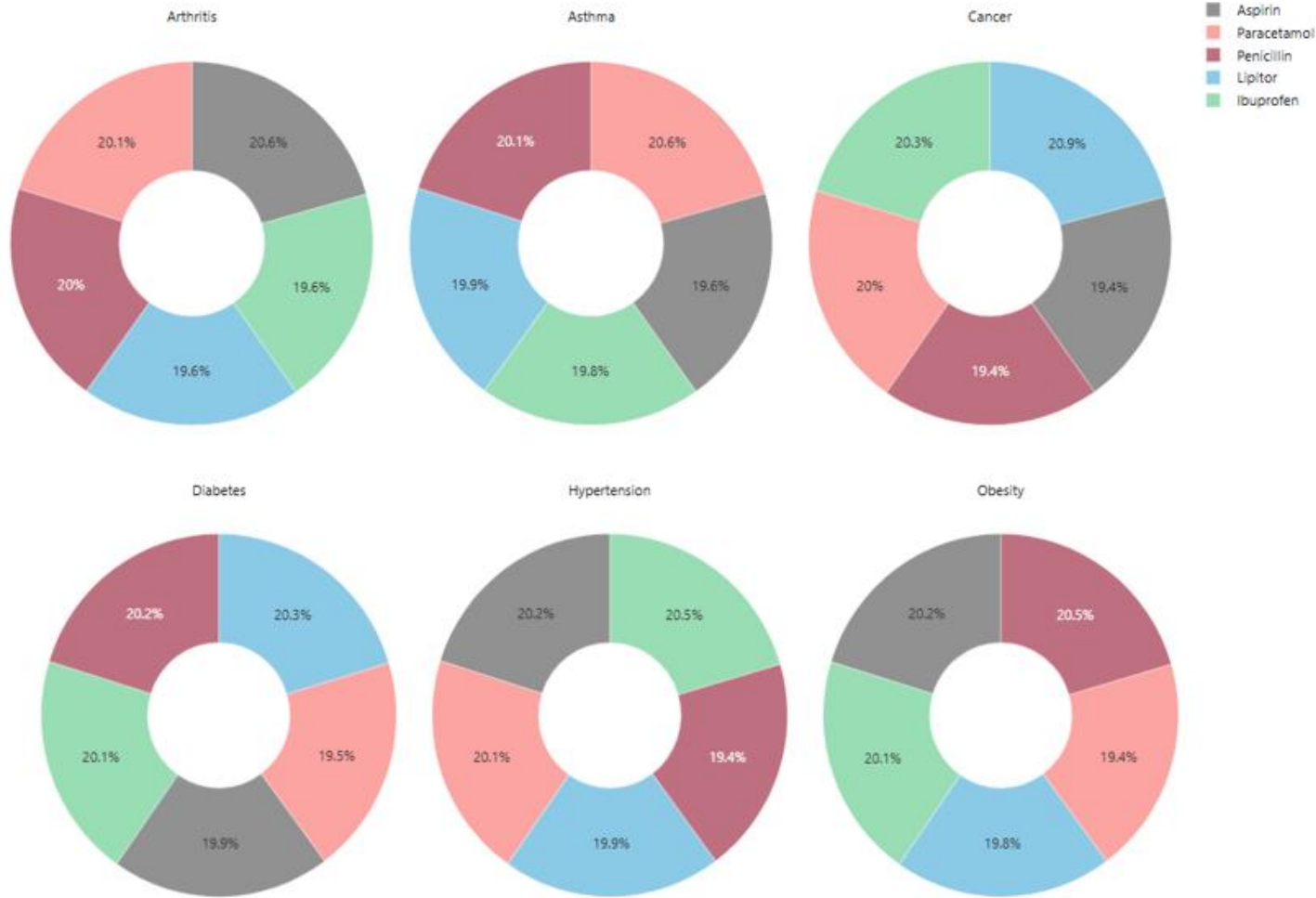
3. Billing Patterns



Average Billing Amount as per each Medical Condition

- The top 5 medical conditions based on average cost of billing are tightly bunched with Obesity being the highest at 25,859.22 and Arthritis the lowest at 25,542.90.
- This suggests that these chronic conditions have equally high treatment costs.
- The close range suggests that treating any one of these conditions is expensive for patients.

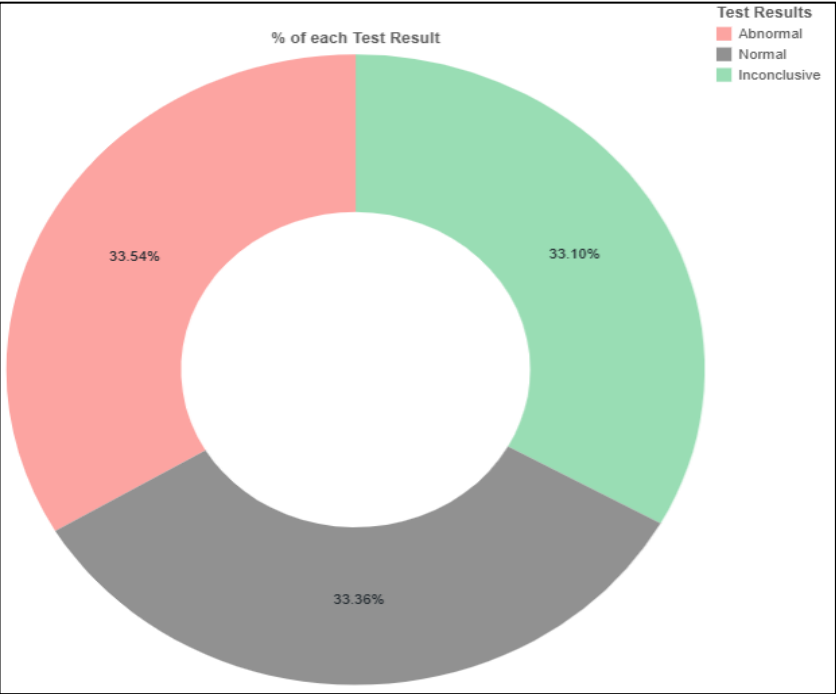
4. Medication Patterns



Medications as per the Medical Conditions

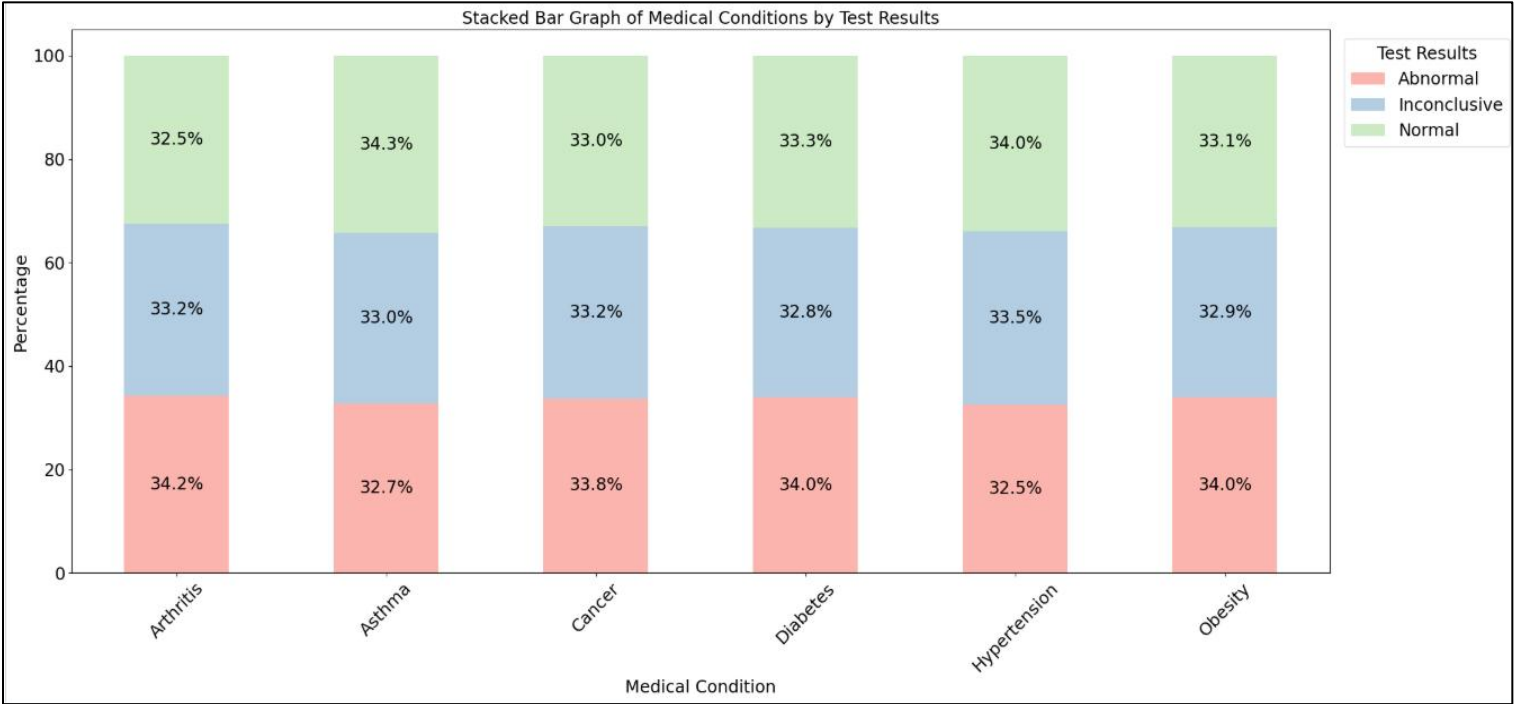
- Aspirin is most used for Arthritis (20.6%) and Asthma (20.6%), showing its relevance in pain and inflammation management.
- Paracetamol is widely used for various conditions, with its highest prevalence in Diabetes (20.2%) and Hypertension (20.1%), probably for alleviating fever and general pain.
- Penicillin is relatively balanced but more prevalent in Obesity (20.5%), suggesting its use in treating infections associated with the condition.
- Lipitor is most used for Cancer (20.9%) and Diabetes (20.3%), possibly for managing cholesterol levels as a comorbidity.
- Ibuprofen shows highest usage in Cancer (20.3%) and Hypertension (20.5%), indicating its importance in inflammation and pain relief across serious conditions

5. Test Results



Test Result distribution

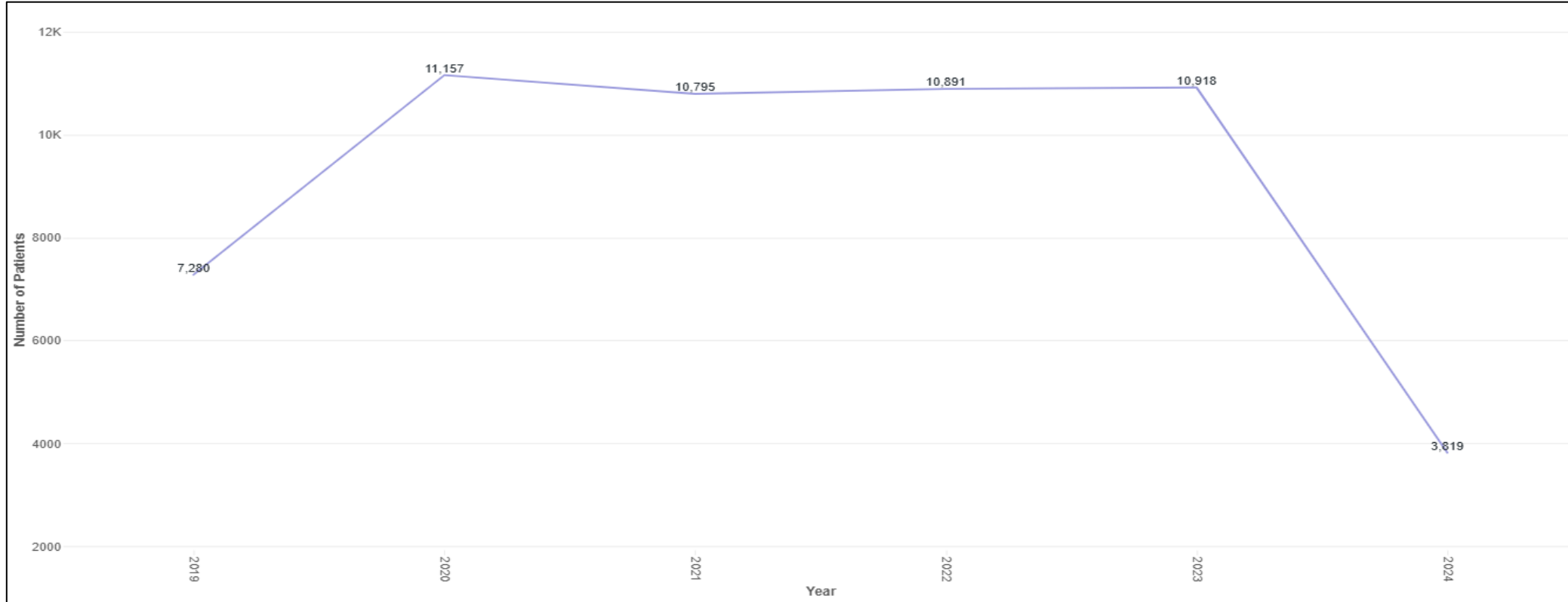
- The test outcomes are nearly evenly spread, with Abnormal results slightly ahead at 33.54%, followed by Normal at 33.36% and Inconclusive at 33.10%.
- This nearly balanced distribution indicates a varied patient population and emphasizes the necessity of follow-up testing and precise diagnostics.



Test Result distribution as per medical condition

- Arthritis (34.2%), Cancer (33.8%), and Obesity (34.0%) show the highest rates of abnormal test results, indicating increased diagnostic issues in these diseases.
- Hypertension (33.5%) and Diabetes (32.8%) exhibit elevated levels of inconclusive outcomes, suggesting the possible challenges in verifying these conditions via initial assessments.
- Asthma (34.3%) and Hypertension (34.0%) exhibit the highest proportions of normal results, indicating improved management or reduced diagnostic ambiguity in these conditions.

6. Trend Analysis



Number of Patients admitted in each year

- The patient count rose markedly from 7,290 in 2019 to a high of 11,157 in 2020, then experienced a minor decrease, stabilizing around 10,800 patients from 2021 through 2023.
- 2024 experienced a steep decrease to 3,819, signaling a significant reduction in patient visits or admissions, potentially caused by policy alterations, service interruptions, or changes in healthcare accessibility.

SOME INTERESTING FINDINGS FROM THE DATASET:

- While diabetes and obesity dominate urgent admissions, cancer leads in emergency cases, and hypertension is mostly elective, suggesting that treatment urgency aligns closely with the nature of the disease rather than just its severity.
- The overall condition spread remained balanced year-to-year, but the rising trends in Diabetes and Asthma highlight shifting healthcare priorities.
- Although obesity has the greatest average billing cost as compared to other medical condition, its proportion of normal test outcomes (33.1%) is notably lower than that of conditions such as asthma or hypertension.
- Simultaneously, diabetes while being prevalent and expensive, is associated with the highest rate of abnormal tests (34%), particularly in emergency admissions. This implies that situations with elevated treatment expenses are not inherently more clinically stable—in reality, they might reflect increased severity and complexity. Thus, making them essential focuses for both clinical intervention and insurance coverage improvement.